

12. The Fourier transform and its applications

So far our methods have been mainly geometric combined with measure theory. We now introduce a very effective analytic tool, the Fourier transform. It comes up naturally in connection with capacities and Hausdorff dimension, because the s -energy of a Radon measure μ with compact support can be written by the formula $I_s(\mu) = c \int |x|^{s-n} |\widehat{\mu}(x)|^2 dx$ in terms of the Fourier transform. In some of the applications the Fourier transform gives an alternative proof for a geometric one, but there are also several results which have only been proven with the help of the Fourier transform. We shall discuss some of them below and some in the next chapter.

Basic formulas

We begin by collecting the basic information about Fourier transforms in $\mathbf{R}^n$, which can be found for example in the book of Stein and Weiss [1].

The Fourier transform $\widehat{f}$ of a Lebesgue integrable function f on $\mathbf{R}^n$ (which may be complex- or extended-real-valued) is defined by

$$\widehat{f}(x) = \int e^{-ix \cdot y} f(y) dy.$$

The basic formulas are the convolution formula

$$(12.1) \quad (f * g)^\wedge = \widehat{f} \widehat{g},$$

the product formula

$$(12.2) \quad \int \widehat{f} g d\mathcal{L}^n = \int f \widehat{g} d\mathcal{L}^n,$$

the Parseval formula

$$(12.3) \quad \int f \overline{g} d\mathcal{L}^n = (2\pi)^{-n} \int \widehat{f} \overline{\widehat{g}} d\mathcal{L}^n,$$

giving the Plancherel formula when $f = g$,

$$(12.4) \quad \int |f|^2 d\mathcal{L}^n = (2\pi)^{-n} \int |\widehat{f}|^2 d\mathcal{L}^n.$$

Here $\bar{z}$ denotes the complex conjugate of the complex number z . These formulas hold in various generalities discussed in Stein and Weiss [1]. They all hold if f and g are infinitely differentiable with compact support and from that they can be extended, with suitable assumptions, to functions in L^p -spaces, measures, and distributions.

There is a very useful formula for the Fourier transform of a radial function. To write it down we need a Bessel function. We shall denote by J_m the Bessel function of the first kind of order m , see Stein and Weiss [1, § 4.3] or G. N. Watson [1, 3.1–40] for its definition and properties. (Here m will be of the form $k/2$ for some non-negative integer k .) Below we shall only list what we are going to need. But first let us note that in $\mathbf{R}^n$ our basic Bessel function will be $J_{(n-2)/2}$, and this is very simple when $n = 3$. Namely, for $t > 0$ (we only need to consider $J_m(t)$ for $t > 0$),

$$J_{1/2}(t) = \sqrt{\frac{2}{\pi}} \frac{\sin t}{\sqrt{t}}.$$

In fact, the asymptotic behaviour of the J_m 's at infinity will be the most relevant for us, and in that respect all J_m 's behave much like $J_{1/2}$: for large t , $J_m(t)$ equals a trigonometric term divided by a square root plus an error term bounded by $C|t|^{-3/2}$.

We shall make use of the following properties of the Bessel functions, which are valid for $0 < t < \infty$.

$$(12.5) \quad |J_m(t)| \leq ct^{-1/2},$$

$$(12.6) \quad |J_m(t)| \leq ct^m,$$

$$(12.7) \quad \frac{d}{dt}(t^m J_m(t)) = t^m J_{m-1}(t).$$

Here c is an absolute constant.

Let g be an $\mathcal{L}^n$ integrable radial function on $\mathbf{R}^n$; $g(x) = \varphi(|x|)$. Then the Fourier transform of g is given by, see Stein and Weiss [1, p. 155],

$$(12.8) \quad \widehat{g}(x) = c|x|^{-(n-2)/2} \int_0^\infty \varphi(s) J_{(n-2)/2}(|x|s) s^{n/2} ds.$$

Finally we shall also need a formula for the Fourier transform of the Riesz kernel

$$k_s: \mathbf{R}^n \rightarrow \overline{\mathbf{R}}, \quad k_s(x) = |x|^{-s},$$

for $0 < s < n$. There is a problem on what that should mean since k_s is not integrable, and not even in any L^p . Motivated by the Parseval

formula (12.3), the Fourier transform $\widehat{g}$ of a function g can often be defined in the distributional sense by requiring that

$$(12.9) \quad \int g \overline{\varphi} d\mathcal{L}^n = (2\pi)^{-n} \int \widehat{g} \overline{\widehat{\varphi}} d\mathcal{L}^n$$

for all $\varphi \in C_0^\infty(\mathbf{R}^n)$. It takes some work to show that for $g = k_s$ such a function $\widehat{k}_s$ exists and is given by (see Landkof [1, (1.1.1)] or Stein [1, V.1.1])

$$(12.10) \quad \widehat{k}_s = c(s, n) k_{n-s}.$$

(When $n = 1$, this is rather simple.)

We shall now study Fourier transforms of measures. For a finite Borel measure μ its Fourier transform is defined by

$$\widehat{\mu}(x) = \int e^{-ix \cdot y} d\mu y.$$

Then $\widehat{\mu}$ is a bounded uniformly continuous function. The formulas (12.1–3) are easily seen to remain valid if $f \in C_0^\infty(\mathbf{R}^n)$ and g (or rather the measure $g d\mathcal{L}^n$) is replaced by a compactly supported Radon measure μ . In order to see this, let $\{\varphi_\varepsilon\}_{\varepsilon>0}$ be an approximate identity as in Chapter 1 and recall from Theorem 1.26 that $\mu * \varphi_\varepsilon \rightarrow \mu$ weakly as $\varepsilon \downarrow 0$. This gives

$$\lim_{\varepsilon \downarrow 0} (\mu * \varphi_\varepsilon)^\wedge(x) = \widehat{\mu}(x) \quad \text{for } x \in \mathbf{R}^n.$$

Note also that

$$|(\mu * \varphi_\varepsilon)^\wedge(x)| \leq \mu(\mathbf{R}^n) < \infty \quad \text{for } x \in \mathbf{R}^n, \varepsilon > 0.$$

We shall now prove two lemmas on which our applications of the Fourier transform to the Hausdorff dimension will be based.

12.11. Lemma. *Let g be a non-negative, locally $\mathcal{L}^n$ integrable, lower semicontinuous function on $\mathbf{R}^n$. If the Fourier transform $\widehat{g}$ of g exists in the sense of (12.9), then for any Radon measure μ on $\mathbf{R}^n$ with compact support,*

$$\iint g(x - y) d\mu x d\mu y \leq (2\pi)^{-n} \int |\widehat{g}| |\widehat{\mu}|^2 d\mathcal{L}^n.$$

Proof. Let $\{\varphi_\varepsilon\}_{\varepsilon>0}$ be an approximate identity such that $\varphi_\varepsilon(x) = \varepsilon^{-n}\varphi(x/\varepsilon)$ with $\text{spt } \varphi \subset B(1/2)$. Then also $\{\psi_\varepsilon\}_{\varepsilon>0}$ with $\psi_\varepsilon = \varphi_\varepsilon * \varphi_\varepsilon$ is an approximate identity and thus by Lemma 1.27

$$(1) \quad \iint g(x-y) d\mu x d\mu y \leq \liminf_{\varepsilon \downarrow 0} \iint (g * \psi_\varepsilon)(x-y) d\mu x d\mu y.$$

Define the Radon measure $\tilde{\mu}$ by

$$\int \varphi(x) d\tilde{\mu} x = \int \varphi(-x) d\mu x \quad \text{for } \varphi \in C_0(\mathbf{R}^n).$$

Then $\widehat{\tilde{\mu}}(x) = \overline{\widehat{\mu}(x)}$.

Several applications of Fubini's theorem and change of variable show that

$$(2) \quad \iint (g * \psi_\varepsilon)(x-y) d\mu x d\mu y = \int \tilde{g}(\mu_\varepsilon * \tilde{\mu}_\varepsilon) d\mathcal{L}^n,$$

where $\tilde{g}(x) = g(-x)$, $\mu_\varepsilon = \varphi_\varepsilon * \mu$ and $\tilde{\mu}_\varepsilon = \varphi_\varepsilon * \tilde{\mu}$. Using (12.9) and (12.1) (also for μ), we compute

$$\begin{aligned} \int \tilde{g}(\mu_\varepsilon * \tilde{\mu}_\varepsilon) d\mathcal{L}^n &= (2\pi)^{-n} \int \overline{\widehat{\tilde{g}}(\mu_\varepsilon * \tilde{\mu}_\varepsilon)} d\mathcal{L}^n \\ &= (2\pi)^{-n} \int \widehat{\tilde{g}} \widehat{\varphi_\varepsilon^2} |\widehat{\mu}|^2 d\mathcal{L}^n \leq (2\pi)^{-n} \int |\widehat{g}| |\widehat{\mu}|^2 |\widehat{\varphi_\varepsilon}|^2 d\mathcal{L}^n. \end{aligned}$$

Since $\widehat{\varphi_\varepsilon}(x) = \widehat{\varphi}(\varepsilon x) \rightarrow \widehat{\varphi}(0) = 1$ uniformly on compact sets as $\varepsilon \downarrow 0$, the right hand side tends to $(2\pi)^{-n} \int |\widehat{g}| |\widehat{\mu}|^2 d\mathcal{L}^n$. Hence the lemma follows from (1) and (2). $\square$

The Fourier transform and energies

We shall now show that equality holds in the previous lemma for $g = k_s$.

12.12. Lemma. *Let μ be a Radon measure on $\mathbf{R}^n$ with compact support. Then for $0 < s < n$,*

$$I_s(\mu) = (2\pi)^{-n} c(s, n) \int |x|^{s-n} |\widehat{\mu}(x)|^2 dx,$$

where $c(s, n)$ is the constant of (12.10).

Proof. Since $\widehat{k_s} \geq 0$ an inspection of the proof of Lemma 12.11 reveals that all we have to do is to show that equality holds in (1) provided $g = k_s$ and $\psi_\varepsilon(x) = \varepsilon^{-n}\psi(x/\varepsilon)$ where ψ is a non-negative function in $C_0^\infty(\mathbf{R}^n)$ with $\int \psi d\mathcal{L}^n = 1$ and $\text{spt } \psi \subset B(1)$. The last inequality in the proof of Lemma 12.11 holds as equality if we choose φ so that $\widehat{\varphi}$ is real-valued. This is true if $\varphi(-x) = \varphi(x)$ for $x \in \mathbf{R}^n$.

Thus we need to show that

$$(1) \quad \limsup_{\varepsilon \downarrow 0} \iint (k_s * \psi_\varepsilon)(x - y) d\mu x d\mu y \leq I_s(\mu).$$

Changing variable,

$$(k_s * \psi_\varepsilon)(z) = \int |z - \varepsilon u|^{-s} \psi(u) du.$$

The function k_s is continuously differentiable in $\mathbf{R}^n \setminus \{0\}$. Thus there exists a constant c_1 such that

$$|\theta - w|^{-s} \leq 1 + c_1|w| \quad \text{for } \theta \in S^{n-1}, w \in B(1/2).$$

Hence for $0 < \varepsilon < 1/4$, $z \in \mathbf{R}^n \setminus \{0\}$ and $u \in B(1)$ with $\sqrt{\varepsilon}|u| \leq |z|$,

$$\begin{aligned} |z - \varepsilon u|^{-s} &= |z|^{-s} |z / |z| - \varepsilon u / |z| |^{-s} \\ &\leq |z|^{-s} (1 + c_1 \varepsilon |u| / |z|) \leq |z|^{-s} (1 + c_1 \sqrt{\varepsilon}). \end{aligned}$$

Thus we obtain

$$\begin{aligned} &\iint (k_s * \psi_\varepsilon)(x - y) d\mu x d\mu y \\ &\leq (1 + c_1 \sqrt{\varepsilon}) I_s(\mu) + \iint \int_{\{u: \sqrt{\varepsilon}|u| > |x-y|\}} |x - y - \varepsilon u|^{-s} \psi(u) du d\mu x d\mu y \\ &\leq (1 + c_1 \sqrt{\varepsilon}) I_s(\mu) + c_2 \iint \int_{\{x: |x-y| < \sqrt{\varepsilon}\}} |x - y|^{-s} d\mu x d\mu y \end{aligned}$$

where c_2 is independent of ε . The last estimate follows from the inequality

$$\int_{B(1)} |z - \varepsilon u|^{-s} du \leq c_3 |z|^{-s},$$

whose proof we leave as an exercise. The last sum tends to $I_s(\mu)$ as $\varepsilon \downarrow 0$, and the lemma is proven. $\square$

We shall now follow Falconer [6] to give some applications of the Fourier transform to the Hausdorff dimension. Other applications will be discussed in the next chapter.

12.13. Lemma. Let μ be a Radon measure on $\mathbf{R}^n$, $n \geq 2$, with compact support. Then for $0 < \varepsilon \leq r < \infty$ and $0 < t \leq 1$,

$$\mu \times \mu(\{(x, y) : r < |x - y| < r + \varepsilon\}) \leq cr^{(n-1)/2} \varepsilon^t I_{t+(n-1)/2}(\mu),$$

where c is a constant depending only on n and t .

Proof. Let g be the characteristic function of the annulus $\{z \in \mathbf{R}^n : r < |z| < r + \varepsilon\}$. Then g is radial and we can use the formula (12.8) to write

$$\widehat{g}(x) = c_1 |x|^{-n} \int_{r|x|}^{(r+\varepsilon)|x|} J_{(n-2)/2}(s) s^{n/2} ds.$$

Here and below $c_1, c_2, \dots$ will denote constants depending only on n . Estimating $J_{(n-2)/2}$ by (12.5), we get

$$|\widehat{g}(x)| \leq c_2 |x|^{-n} \int_{r|x|}^{(r+\varepsilon)|x|} s^{(n-1)/2} ds \leq c_3 r^{(n-1)/2} |x|^{(1-n)/2} \varepsilon.$$

Using (12.7) and (12.5) we also have the estimate

$$\begin{aligned} |\widehat{g}(x)| &= \left| c_1 |x|^{-n} \int_{r|x|}^{(r+\varepsilon)|x|} \frac{d}{ds} (s^{n/2} J_{n/2}(s)) ds \right| \\ &= \left| c_1 |x|^{-n} ((r+\varepsilon)|x|)^{n/2} J_{n/2}((r+\varepsilon)|x|) - (r|x|)^{n/2} J_{n/2}(r|x|) \right| \\ &\leq c_4 r^{(n-1)/2} |x|^{-(n+1)/2}. \end{aligned}$$

With the help of these inequalities and Lemmas 12.11 and 12.12 we

obtain

$$\begin{aligned}
 & \mu \times \mu(\{(x, y) : r < |x - y| < r + \varepsilon\}) \\
 &= \iint g(x - y) d\mu x d\mu y \\
 &\leq (2\pi)^{-n} \int |\widehat{g}| |\widehat{\mu}|^2 d\mathcal{L}^n \\
 &\leq c_5 r^{(n-1)/2} \left[\varepsilon \int_{\{|x| \leq 1/\varepsilon\}} |x|^{(1-n)/2} |\widehat{\mu}(x)|^2 dx \right. \\
 &\quad \left. + \int_{\{|x| > 1/\varepsilon\}} |x|^{-(n+1)/2} |\widehat{\mu}(x)|^2 dx \right] \\
 &\leq c_5 \varepsilon^t r^{(n-1)/2} \left[\int_{\{|x| \leq 1/\varepsilon\}} |x|^{(1-n)/2+t-1} |\widehat{\mu}(x)|^2 dx \right. \\
 &\quad \left. + \int_{\{|x| > 1/\varepsilon\}} |x|^{-(n+1)/2+t} |\widehat{\mu}(x)|^2 dx \right] \\
 &= c_5 \varepsilon^t r^{(n-1)/2} \int |x|^{(n-1)/2+t-n} |\widehat{\mu}(x)|^2 dx \\
 &= c \varepsilon^t r^{(n-1)/2} I_{t+(n-1)/2}(\mu).
 \end{aligned}$$

□

Distance sets

We now apply the above lemma to get information about the Hausdorff dimension of distance sets. The *distance set* of a subset A of $\mathbf{R}^n$ is

$$D(A) = \{|x - y| : x, y \in A\}.$$

Recall from Exercise 2.4 that $D(A)$ contains some interval $[0, \varepsilon]$, $\varepsilon > 0$, if A is $\mathcal{L}^n$ measurable and $\mathcal{L}^n(A) > 0$. But what if A only has Hausdorff dimension close to n instead of positive Lebesgue measure? The porous sets of 4.12 can easily be used to show that then, even if $\dim A = n$, the distance set $D(A)$ need not contain any interval with left end-point at 0. However, it is not known whether $D(A)$ must contain some other non-degenerate intervals if $\dim A$ is sufficiently close to n and A is a Borel set. The following result of Falconer tells us that at least $D(A)$ has positive Lebesgue measure if $\dim A > (n + 1)/2$.

12.14. Theorem. Let A be a Borel set in $\mathbf{R}^n$.

- (1) If $\dim A > (n+1)/2$, then $\mathcal{L}^1(D(A)) > 0$.
- (2) If $(n-1)/2 \leq \dim A \leq (n+1)/2$, then $\dim D(A) \geq \dim A - (n-1)/2$.

Proof. This theorem has really no content in $\mathbf{R}^1$, so we may assume $n \geq 2$. Clearly we may also assume that A is bounded, say $d(A) \leq R$. Then $D(A) \subset [0, R]$. Finally, we may assume $\dim A > (n-1)/2$, and we choose $t \leq 1$ such that $0 < t < \dim A - (n-1)/2$. Then by Theorem 8.9 there is a Radon measure μ with $\text{spt } \mu \subset A$, $\mu(A) = 1$ and $I_{t+(n-1)/2}(\mu) < \infty$.

Suppose that the open intervals $(r_i, r_i + \varepsilon_i)$ cover $D(A) \setminus \{0\}$ and that $0 < \varepsilon_i \leq r_i \leq R$. Then

$$(A \times A) \setminus \{(x, y) : x = y\} \subset \bigcup_i \{(x, y) : r_i < |x - y| < r_i + \varepsilon_i\}.$$

Since $I_{t+(n-1)/2}(\mu) < \infty$, the singletons have μ measure zero and so the diagonal $\{(x, y) : x = y\}$ has $\mu \times \mu$ measure zero. Thus by Lemma 12.13

$$\begin{aligned} 1 &= \mu \times \mu(A \times A) \leq \sum \mu \times \mu(\{(x, y) : r_i < |x - y| < r_i + \varepsilon_i\}) \\ &\leq cR^{(n-1)/2} I_{t+(n-1)/2}(\mu) \sum_i \varepsilon_i^t. \end{aligned}$$

If $\dim A > (n+1)/2$, we can take $t = 1$ to obtain

$$\mathcal{L}^1(D(A)) \geq [cR^{(n-1)/2} I_{(n+1)/2}(\mu)]^{-1} > 0.$$

In the second case we have similarly $\mathcal{H}^t(D(A)) > 0$. □

Borel subrings of $\mathbf{R}$

As a corollary we obtain some information about the possible dimensions of the subrings of the reals.

12.15. Theorem. Let R be a subring of the ordinary ring of the real numbers. If R is a Borel set, then either $0 \leq \dim R \leq 1/2$ or $\dim R = 1$.

Proof. Since R is a ring, the set

$$D^2(R \times R) = \{|x - y|^2 : x, y \in R \times R\}$$

is contained in R . The map $t \mapsto t^2$ preserves the Hausdorff dimension, whence $\dim D^2(R \times R) = \dim D(R \times R)$. By Theorem 8.10 $\dim(R \times R) \geq 2 \dim R$. Using these facts and Theorem 12.14, we obtain

$$\begin{aligned} \dim R &\geq \dim D^2(R \times R) = \dim D(R \times R) \\ &\geq \min \{1, \dim(R \times R) - 1/2\} \\ &\geq \min \{1, 2 \dim R - 1/2\}. \end{aligned}$$

If the minimum equals 1, we have $\dim R = 1$. Otherwise, $\dim R \geq 2 \dim R - 1/2$, which gives $\dim R \leq 1/2$. $\square$

Several remarks are now in order:

12.16. Remarks. (1) As noted above, Theorem 12.14 does not say anything if $n = 1$. And there is really nothing to say. One can construct compact sets B and C in $\mathbf{R}$ such that $\dim B = 0$, $\dim D(B) = 1$, and $\dim C = \dim D(C)$ equals any given number between zero and one. That is, the trivial bounds $\dim A \leq \dim D(A) \leq 1$ are the best possible.

(2) In higher dimensions $n \geq 2$ the best possible bounds in the situation of Theorem 12.14 are not known. Probably $\dim A > n/2$ could be enough for $\mathcal{L}^1(D(A)) > 0$. This would be sharp as shown by an example in Falconer [6]. Sharpenings in some cases have been obtained in Hawkes [2] and Mattila [13]. A very recent progress was achieved by Bourgain [4] who used his profound Fourier-analytic techniques from Bourgain [3] to improve the bound $(n+1)/2$ for $n = 2$ and $n = 3$. For example, he showed that $\mathcal{L}^1(D(A)) > 0$ holds if A is a Borel set in $\mathbf{R}^2$ with $\dim A > 13/9$. His method consists of showing that the image of $\mu \times \mu$ under the distance map is in L^2 if $I_s(\mu) < \infty$ for s sufficiently large; see Exercise 2. The properties of the difference sets of specific Cantor sets can be very intricate. A recent extensive study of them is Larsson [2], see also Larsson [1] and Sannami [1].

(3) Also the sharpness of Theorem 12.15 is unknown; it may be that any Borel subring of the reals must have Hausdorff dimension 0 or 1. The problem is also open for the subfields. For subgroups this is not true: Erdős and Volkmann [1], see also Falconer [16, § 12.4], have shown that there exist Borel subgroups of $\mathbf{R}$ of all dimensions between 0 and 1.

(4) The above results for the distance sets and rings hold more generally for Suslin sets, but they are false for arbitrary subsets of $\mathbf{R}$. R. O. Davies (unpublished) has constructed using the continuum hypothesis non-Suslin subrings of the reals of all dimensions in $[0, 1]$. Similarly Falconer [6] has shown that there are only some rather straightforward inequalities for the dimensions of the distance sets of arbitrary subsets of $\mathbf{R}^n$.

(5) No proofs for Theorems 12.14 and 12.15 are known without the Fourier transform. Falconer [5] gave another proof for Theorem 12.15 using orthogonal projections but that too relied on the Fourier transform.

Fourier dimension and Salem sets

12.17. The formula in Lemma 12.12 for $I_s(\mu)$ and the relations between Hausdorff dimension and capacities discussed in Chapter 8 show that the Hausdorff dimension of a Borel set A in $\mathbf{R}^n$ can be determined by looking at the behaviour of Fourier transforms of measures supported by A . More precisely, $\dim A$ equals the supremum of the numbers s such that there exists a Radon measure μ with $\text{spt } \mu \subset A$, $\mu(A) = 1$ and $\int |x|^{s-n} |\hat{\mu}(x)|^2 dx < \infty$. The finiteness of $\int |x|^{s-n} |\hat{\mu}(x)|^2 dx$ tells us that for “most points” x with large norm,

$$|\hat{\mu}(x)| \leq c|x|^{-s/2}.$$

However, we cannot usually require this for all x . In fact, such a requirement leads to another dimension, called the *Fourier dimension*, $\dim_F A$, of A . By the definition, $\dim_F A$ is the unique number in $[0, n]$ such that for any $0 < s < \dim_F A$ there exists a non-zero Radon measure μ with $\text{spt } \mu \subset A$ and $|\hat{\mu}(x)| \leq |x|^{-s/2}$ for $x \in \mathbf{R}^n$, and that for $\dim_F A < s < n$ no such measure exists.

We have for any Borel set $A \subset \mathbf{R}^n$,

$$\dim_F A \leq \dim A.$$

The inequality is often strict. For example, $\dim_F C = 0$ for the Cantor ternary set $C = C(1/3)$ in 4.10; in fact, C supports no non-zero Radon measure whose Fourier transform would tend to zero at infinity, see Kahane and Salem [1]. Sets A for which $\dim_F A = \dim A$ are called *Salem sets*. They are very rare as deterministic sets, some non-trivial ones were constructed by Kahane [2] and by Kaufman [5]. However, as random sets they abound. The first constructions by Salem [1] were random, and more recently Kahane has shown that many really significant random sets are Salem sets, see Kahane [3]. Thus for example if $\omega: [0, \infty) \rightarrow \mathbf{R}^n$ denotes the n -dimensional Brownian motion, then for any compact set $F \subset [0, \infty]$ the image $\omega(F)$ is almost surely a Salem set. That is, recall Theorem 9.18,

$$\dim_F \omega(F) = \dim \omega(F) = \min\{n, 2 \dim F\}.$$

For further discussions on these topics and also for questions related to Fourier series, see the books Kahane and Salem [1] and Kahane [3].

For Salem sets one can improve the previous results on distance sets. In Mattila [13] it was shown that if a Borel subset A of $\mathbf{R}^n$, $n \geq 1$, is a Salem set, then $\mathcal{L}^1(D(A)) > 0$, provided $\dim A > n/2$, and $\dim D(A) \geq 2 \dim A - n + 1$, provided $\dim A \leq n/2$. See also Hawkes [2].

Although $\hat{\mu}$ need not go to zero at infinity for Radon measures μ with finite s -energy, the averages

$$\sigma(\mu)(r) = \int_{S^{n-1}} |\hat{\mu}(r\zeta)|^2 d\mathcal{H}^{n-1}\zeta$$

over spheres $S(r)$ do tend to zero as $r \rightarrow \infty$ if $I_s(\mu) < \infty$ for some $0 < s < n$, $n \geq 2$. Moreover,

$$(1) \quad \sigma(\mu)(r) \leq cr^{-s} I_s(\mu) \quad \text{for } r > 0,$$

provided $0 < s \leq (n-1)/2$. The decay r^{-s} is the best possible one can get from the information $I_s(\mu) < \infty$, cf. Exercise 7. The estimate (1) was derived in Mattila [13]. It was also shown there for $n = 2$ that (1) fails for $s > (n-1)/2$ and some weaker estimates in $\mathbf{R}^n$ when $s > (n-1)/2$ were obtained. Later Sjölin [1] improved these estimates and constructed counterexamples in $\mathbf{R}^n$. Still the best exponent $t(s)$ such that $I_s(\mu) < \infty$ implies $\sigma(\mu)(r) < r^{-t(s)}$ for large r is not known for all s . Information on this can also be obtained from Bourgain [4].

Kaufman [3] studied the decay at infinity along lines through the origin of the Fourier transforms of Radon measures μ on $\mathbf{R}^2$. He showed that if $I_1(\mu) < \infty$ then $\hat{\mu}$ tends to zero along $\gamma_{2,1}$ almost all $L \in G(2,1)$. Moreover, if $I_s(\mu) < \infty$ then the Hausdorff dimension of the exceptional set of lines is at most $2 - s$.

The construction of Riesz products is often an effective Fourier-analytic method of building singular measures with interesting properties. They are defined as weak limits of products of trigonometric functions. For their properties, see Peyrière [1] and Fan [2], and for some particular examples Freedman and Pitman [1] and Kahane [2].

Strichartz [1]–[7] has studied extensively the behaviour of the Fourier transform of self-similar and other fractal measures, see also Hudson and Leckband [1], Lau [1] and Lau and Wang [1].

Exercises.

1. Show that for a finite Borel measure μ with compact support the Fourier transform $\hat{\mu}$ is a Lipschitz function.

2. Let $d: \mathbf{R}^n \times \mathbf{R}^n \rightarrow \mathbf{R}$ be the distance function, $d(x, y) = |x - y|$. Show that if μ is a Radon measure on $\mathbf{R}^n$ with compact support and with $I_{(n+1)/2}(\mu) < \infty$, then $d_{\#}(\mu \times \mu) \ll \mathcal{L}^1$ with bounded Radon–Nikodym derivative. Hint: Use Lemma 12.13.
3. Let $C(\lambda)$, $0 < \lambda < 1/2$, be as in 4.10. Show that $\mathcal{L}^1(D(C(\lambda))) > 0$ if and only if $\lambda \geq 1/3$, and that in this case $D(C(\lambda))$ is an interval.
4. Show that there exist compact sets $A \subset \mathbf{R}^n$ with $\dim A = n$ such that $D(A)$ contains no interval $[0, \varepsilon]$, $\varepsilon > 0$.
5. Prove that $\dim_F A \leq \dim A$ for Borel sets $A \subset \mathbf{R}^n$.
6. If $A \subset \mathbf{R}^n$ with $\dim_F A > 0$, then the group $(G, +)$ generated by A equals $\mathbf{R}^n$. Hint: If $\text{spt } \mu \subset A$, then every k -fold convolution $\mu^k = \mu * \cdots * \mu$ has support in G . For sufficiently large k , $(\mu^k)^\wedge \in L^2$, whence $\mu^k \ll \mathcal{L}^n$. This implies that G contains a closed set with positive Lebesgue measure and the rest follows as in Exercise 2.5.

For a construction of a compact set $A \subset \mathbf{R}$ with $\dim A = 1$ but $G \neq \mathbf{R}$, where G is the group generated by A , see Beck [1].

7. Let μ be a non-zero Radon measure on $\mathbf{R}^n$ with $\dim \text{spt}(\mu) = s$. Show that the estimate

$$\sigma(\mu)(r) \leq r^{-t} \quad \text{for } r > 0$$

can hold only if $t \leq s$.